

Aim: To implement a weather application using Flask that retrieves and displays weather information based on user input.

Q1. What is Flask?

Flask is a lightweight and flexible **micro web framework** for Python. It provides tools, libraries, and patterns to build web applications and APIs with minimal code. Unlike full-stack frameworks, Flask keeps the core simple but is highly extensible with extensions for databases, authentication, etc.

Q2. What is an API?

API stands for **Application Programming Interface**. It is a set of rules and protocols that allows different software applications to communicate with each other. In web development, a **Web API** (often `RESTful`) exposes endpoints (URLs) that accept requests (e.g., `GET`, `POST`) and return data (commonly JSON or XML) without a graphical user interface.

Q3. What is the purpose of a virtual environment?

A virtual environment is an isolated Python environment that allows you to manage project-specific dependencies without interfering with system-wide Python packages. Its main purposes are:

- Avoid version conflicts between projects.
- Keep global Python clean and minimal.
- Reproduce the exact environment across development, testing, and production (e.g., via `requirements.txt`).

Q4. What is HTTP GET method?

The `GET` method is one of the HTTP request methods used to **retrieve data** from a server. It appends parameters to the URL (query string) and should be **idempotent** (repeating the same request has the same effect) and **safe** (does not change server state). Examples: loading a webpage, fetching search results, or calling an API endpoint to read data.

Q5. How does a web application handle user input?

- **Capture input** via HTML forms, JavaScript events, or URL parameters.
- **Send to server** using an HTTP request (`GET` with query params, `POST/PUT` with form data or JSON in the request body).
- **Validate and sanitize** on the server to prevent security issues (e.g., SQL injection, XSS).
- **Process the data** (e.g., save to a database, perform calculations).
- **Generate a response** (HTML page, redirect, or JSON data) sent back to the client.
- **Provide feedback** to the user (success/error messages, updated UI).

Q6. Implement a weather application using Flask that retrieves and displays weather information based on user input.

Code (app.py):

```
from flask import Flask, render_template, request
import requests
from datetime import datetime

app = Flask(__name__)

API_KEY = "efa2f42bad1530ef4a494f012912c0ac"
BASE_URL = "https://api.openweathermap.org/data/2.5/weather"

@app.route('/', methods=['GET', 'POST'])
def index():
    weather = None
    error = None

    if request.method == 'POST':
        city = request.form['city']

        params = {
            'q': city,
            'appid': API_KEY,
            'units': 'metric'
        }

        response = requests.get(BASE_URL, params=params)
        data = response.json()

        print(data)

        if str(data.get("cod")) != "200":
            error = data.get("message", "Error fetching data")
        else:
            sunrise_time =
datetime.fromtimestamp(data['sys']['sunrise']).strftime('%H:%M:%S')
            sunset_time =
datetime.fromtimestamp(data['sys']['sunset']).strftime('%H:%M:%S')
            data_time = datetime.fromtimestamp(data['dt']).strftime('%Y-%m-%d %H:%M:%S')

            weather = {
                'temperature': data['main']['temp'],
                'feels_like': data['main']['feels_like'],
                'description': data['weather'][0]['description'],
                'weather_main': data['weather'][0]['main'],
                'humidity': data['main']['humidity'],
                'pressure': data['main']['pressure'],
                'wind': data['wind']['speed'],
                'city': data['name'],
                'country': data['sys']['country'],
                'last_updated': data_time,
                'sunrise': sunrise_time,
                'sunset': sunset_time
            }

    return render_template('index.html', weather=weather, error=error)

if __name__ == '__main__':
    app.run(debug=True)
```

Code (root_layout.html):

```

<!doctype html>
<html>
  <head>
    <title>Weather App</title>
  </head>

  <body>
    <div class="container">
      <div class="search-section">
        <form method="POST"
class="input-group">
          <input
            type="text"
            name="city"
            placeholder="Enter city name (e.g., London, Tokyo, New York)"
            required
          />
          <button
type="submit"><img alt="search icon" data-bbox="218 403 238 418"/> Get Weather</button>
        </form>
      </div>

      <div class="weather-grid">
        {% if error %}
          <div class="error-
message"><img alt="warning icon" data-bbox="175 505 195 520"/> {{ error }}</div>
        {% endif %} {% if weather %}

          <div class="weather-card
city-card">
            <div class="city-
name">{{ weather.city }}</div>
            <div class="country-
name">{{ weather.country }}</div>
            <div class="temp-
large">{{ weather.temperature }}°C</div>
            <div class="card-
description">
              {{
weather.description | capitalize }}
            </div>
          </div>

          <div class="weather-
card">
            <div class="card-
header">
              <span
class="icon"><img alt="cloud with rain icon" data-bbox="208 845 228 860"/></span>
              <h3>Feels
Like</h3>
            </div>
            <div class="card-
value">
              {{
weather.feels_like }}<span class="card-
unit"
              >°C</span>
            </div>
            <div class="card-
description">Real feel temperature</div>
          </div>

          <div class="weather-
card">
            <div class="card-
header">
              <span
class="icon"><img alt="sun with clouds icon" data-bbox="648 375 668 390"/></span>
              <h3>Category</h3>
            </div>
            <div class="card-
value">{{ weather.weather_main }}</div>
            <div class="card-
description">Weather type</div>
          </div>

          <div class="weather-
card">
            <div class="card-
header">
              <span
class="icon"><img alt="water drop icon" data-bbox="648 575 668 590"/></span>
              <h3>Humidity</h3>
            </div>
            <div class="card-
value">
              {{
weather.humidity }}<span class="card-
unit">%</span>
            </div>
            <div class="card-
description">Moisture level</div>
          </div>

          <div class="weather-
card">
            <div class="card-
header">
              <span
class="icon"><img alt="bar chart icon" data-bbox="648 830 668 845"/></span>
              <h3>Pressure</h3>
            </div>
            <div class="card-
value">
              {{
weather.pressure }}<span class="card-
unit">hPa</span>
            </div>
            <div class="card-
description">Atmospheric pressure</div>
          </div>
        </div>
      </div>
    </div>
  </body>
</html>

```

```

        <div class="card-
value">
            {{
weather.pressure }}<span class="card-
unit">hPa</span>
        </div>
        <div class="card-
description">Atmospheric pressure</div>
        </div>

        <div class="weather-
card">
            <div class="card-
header">
                <span
class="icon"><img alt="Wind icon" data-bbox="208 301 223 316"/></span>
                <h3>Wind
Speed</h3>
            </div>
            <div class="card-
value">
                {{ weather.wind
}}<span class="card-unit">m/s</span>
            </div>
            <div class="card-
description">Wind intensity</div>
            </div>

        <div class="weather-
card">
            <div class="card-
header">
                <span
class="icon"><img alt="Sunrise icon" data-bbox="208 558 223 573"/></span>
                <h3>Sunrise</h3>
            </div>
            <div class="card-
value">{{ weather.sunrise }}</div>
            <div class="card-
description">Morning begins</div>

```

```

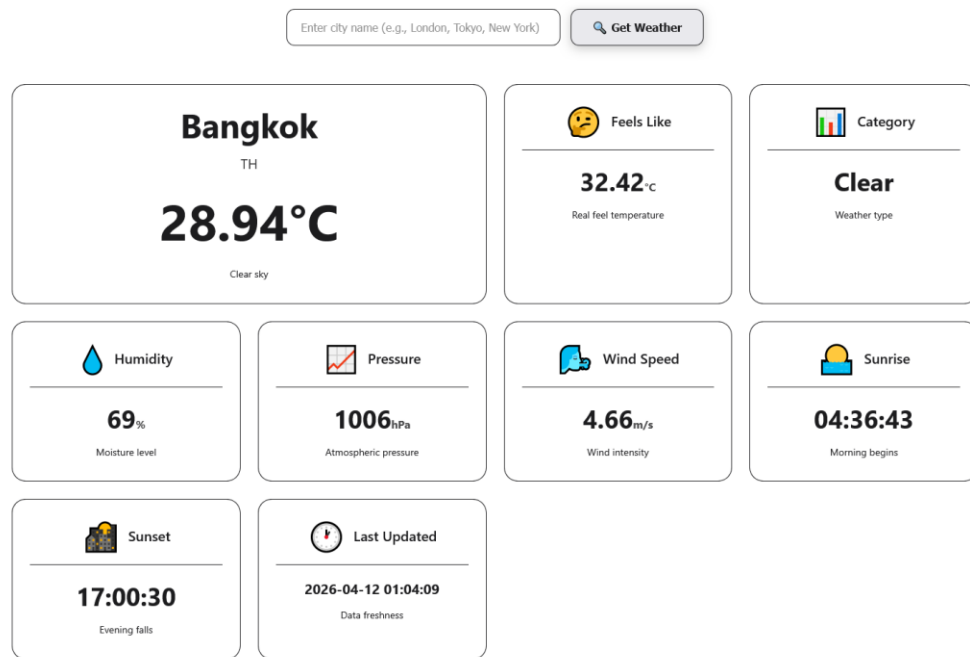
        </div>

        <div class="weather-
card">
            <div class="card-
header">
                <span
class="icon"><img alt="Sunset icon" data-bbox="648 188 663 203"/></span>
                <h3>Sunset</h3>
            </div>
            <div class="card-
value">{{ weather.sunset }}</div>
            <div class="card-
description">Evening falls</div>
            </div>

        <div class="weather-
card">
            <div class="card-
header">
                <span
class="icon"><img alt="Clock icon" data-bbox="648 388 663 403"/></span>
                <h3>Last
Updated</h3>
            </div>
            <div class="card-
value" style="font-size: 20px">
                {{
weather.last_updated }}
            </div>
            <div class="card-
description">Data freshness</div>
            </div>
        {% endif %}
    </div>
</body>
</html>

```

Output:



Conclusion:

Successfully implemented a weather application using Flask that retrieves and displays weather information based on user input.